

Statistical Process Control (SPC)

Duration: 1 Day

Mode: Online / Classroom / On-Site

Certification: QLSS Certificate of Completion



Course Introduction

Statistical Process Control (SPC) is a key quality management technique used to monitor, control, and improve processes through statistical analysis. It enables early detection of process instability and helps reduce variation, waste, and defects.

This training introduces core SPC concepts including control charts, process capability indices, and variation types. Participants learn to apply SPC tools practically on real manufacturing and transactional processes, with special focus on interpreting charts, identifying special causes, and taking corrective action.



Pre-requisites

- Basic understanding of quality or production processes
- Exposure to manufacturing or service data
- No prior statistical background required



Learning Objectives

- Understand the role of SPC within IATF 16949 and APQP
- Differentiate between common and special cause variation
- Construct and interpret control charts ($\bar{X}$ -R, $\bar{X}$ -S, p, np, c, u)
- Calculate and apply Cp, Cpk, Pp, Ppk indices
- Use SPC for continuous improvement and problem-solving
- Select appropriate control charts for different data types
- Integrate SPC with process audits and control plans



Course Agenda

1. Introduction to SPC

- History and need for statistical process control
- Benefits of using SPC in manufacturing and services
- Role of SPC in IATF 16949 / VDA 6.3

2. Variation and Process Behavior

- Common vs. special cause variation
- Stable vs. unstable processes
- Identifying patterns, trends, and shifts

3. Control Charts – Variables Data

- $\bar{X}$ -R and $\bar{X}$ -S Charts: when to use, how to calculate
- Individual-Moving Range (I-MR) Charts
- Control limits vs. specification limits

4. Control Charts – Attribute Data

- p, np, c, u Charts explained with examples
- Data type identification and chart selection
- Real-world use cases in inspection processes

5. Process Capability Analysis

- Cp, Cpk, Pp, Ppk: interpretation and calculation
- Normal distribution check using histogram and normality tests
- Capability study prerequisites and sample size

6. SPC and Continuous Improvement

- Use of SPC in problem-solving (PDCA, DMAIC)
- Linking SPC signals to corrective actions
- Creating SPC-based control plans



What You'll Receive

- QLSS Certificate of Completion
- Training Material / Handout (PDF Format)



Who Should Attend

- Quality Engineers and Supervisors
- Process Owners and Manufacturing Leads
- Production and Operation Team Members
- Internal Auditors and Six Sigma Practitioners
- SPC & Data Analysis Enthusiasts



Trainer Profile

Our SPC trainers are experts in statistical quality techniques with years of practical experience across industries including automotive, electronics, and healthcare. They simplify statistics through visual tools and teach interpretation through real case studies, ensuring that trainees confidently use SPC in day-to-day quality decisions.



Instructional Methodology

- Hands-on control chart plotting
- Use of Excel-based SPC templates
- Group discussion and exercises on chart interpretation
- Calculator-based Cp/Cpk exercises
- Fault detection and response simulation



What Makes QLSS SPC Training Unique

- Visual learning with minimal jargon
- Focused on real-time interpretation and action
- Practical templates for quick deployment
- Workshop-based learning and interaction
- Designed for both production and quality professionals